



SOFTWARE ENGINEERING I PROJECT TECHNICAL DOCUMENTATION

1. Date Submitted: May 9, 2026

2. Project Manager: Zapico, Zaki

3. Members:

Member 1	Roquid, Jenny Lyn
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4. Contact Person

Name of Contact Person:	Rhona Mae Aganan (Owner) & Jimwell Mamangun (Co-Owner)
Position Designation: or	Owner & Co-Owner / Operations Manager
Landline No.:	N/A (Mobile pop-up business)
Mobile No.:	09762340887
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5. The Company

Company Name:	TravelBean MNL
Company Address:	Pampanga St. Tondo, Manila
Major Products/Services:	Coffee beverages (espresso-based, cold brew, specialty drinks)
Minor Products/Services:	Bottled water, event catering setup



6. Nature of the Proposed System

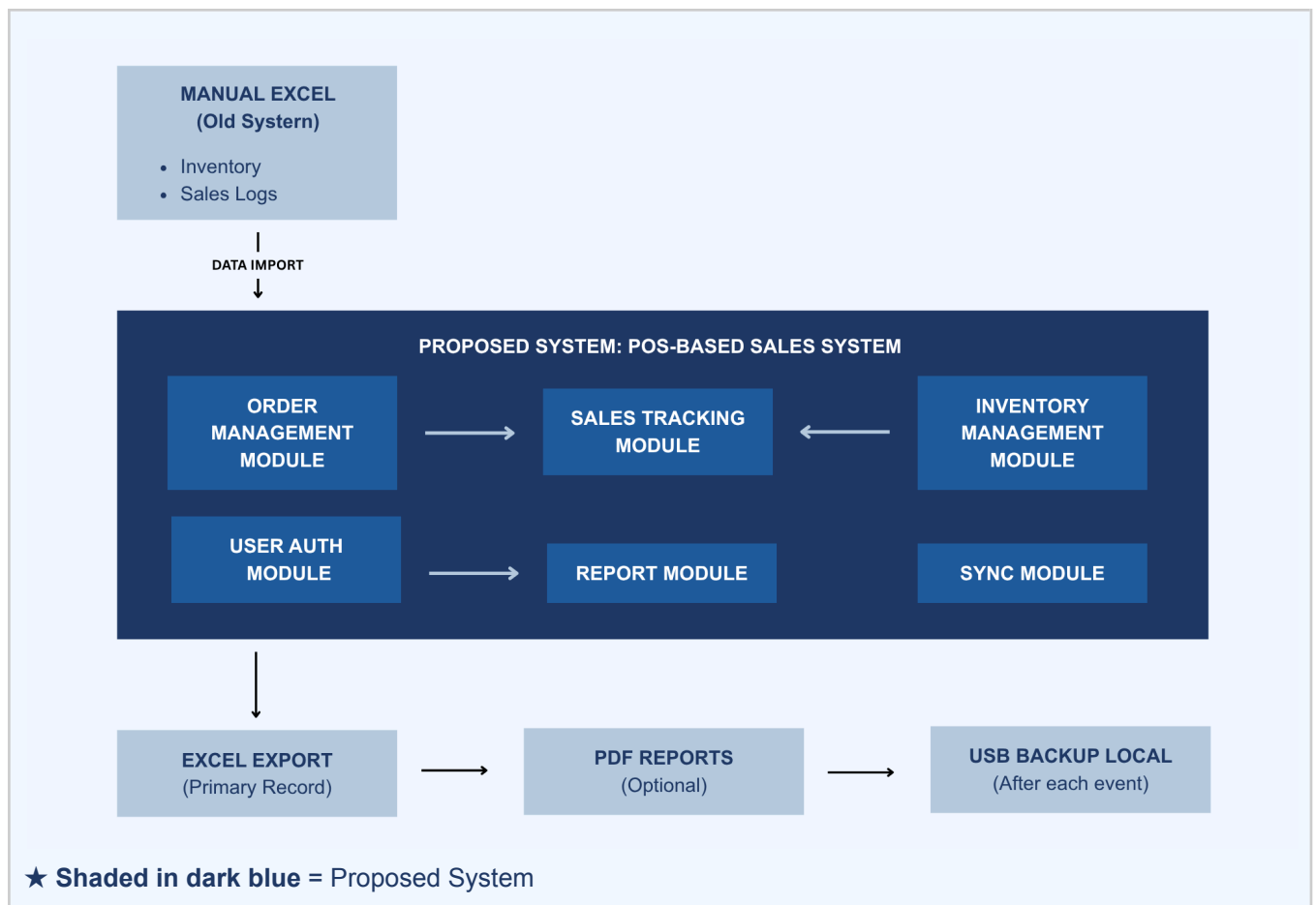
The proposed system is a standalone Point-of-Sale (POS) application designed specifically for TravelBean MNL's mobile pop-up coffee business. It automates sales tracking, order management, and inventory monitoring at events where internet connectivity is unreliable or unavailable.

The system operates offline-first on both a laptop and a tablet. No internet connection is required during events. All data is stored locally on the laptop's MySQL database. The laptop creates its own WiFi hotspot, and tablets connect directly to the laptop. This creates a local network where the tablet accesses the POS system through a web browser using the laptop's local IP address.

When the event venue has available WiFi or mobile hotspot, the owner can choose to sync data between devices or export sales data. However, no cloud backup is used all backups are done locally to a USB drive after each event.

6.1. Interrelationship of the Proposed System

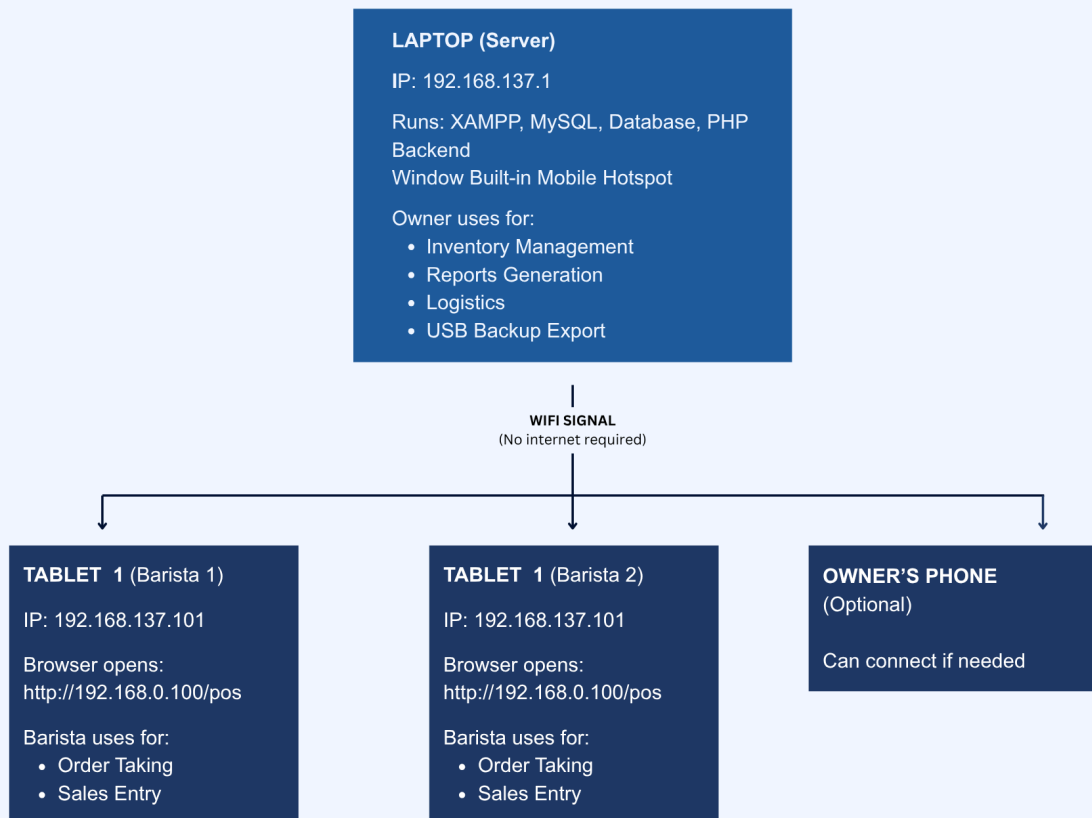
The diagram below illustrates how the POS system interacts with the other systems in TravelBean MNL's operations. The proposed system is highlighted.





6.2. Organizational Chart & Department Using the System

TravelBean MNL is a small mobile pop-up coffee business. The Owner/Co-Owner handles logistics, while the Barista/Staff directly uses the POS system for order-taking and sales.



NETWORK SPECIFICATIONS:

- | | |
|---------------|--|
| a. Bandwidth: | No minimum required during events (offline operation) |
| b. ISP: | NOT REQUIRED for operations. Owner may use mobile data only for optional email reports AFTER events. |
| c. Topology: | Star topology using WiFi hotspot |
| d. Cabling: | No physical cabling - wireless only |
| e. Router: | No dedicated router - laptop hotspot or pocket WiFi |
| f. Switches: | None required |

■ **Dark navy** = Barista/Staff (primary POS users) ■ **Medium blue** = Owner/Co-Owner (Logistics)



6.3. Implementation of the Proposed System

Selected: **Partially Automated with Fully Computerized Transaction Processing**

The system is partially automated with fully computerized transaction processing. Staff manually enter orders into the system. However, all calculations (totals, change), inventory deductions, sales logging, conflict resolution, and report generation are handled digitally with no paper-based steps.

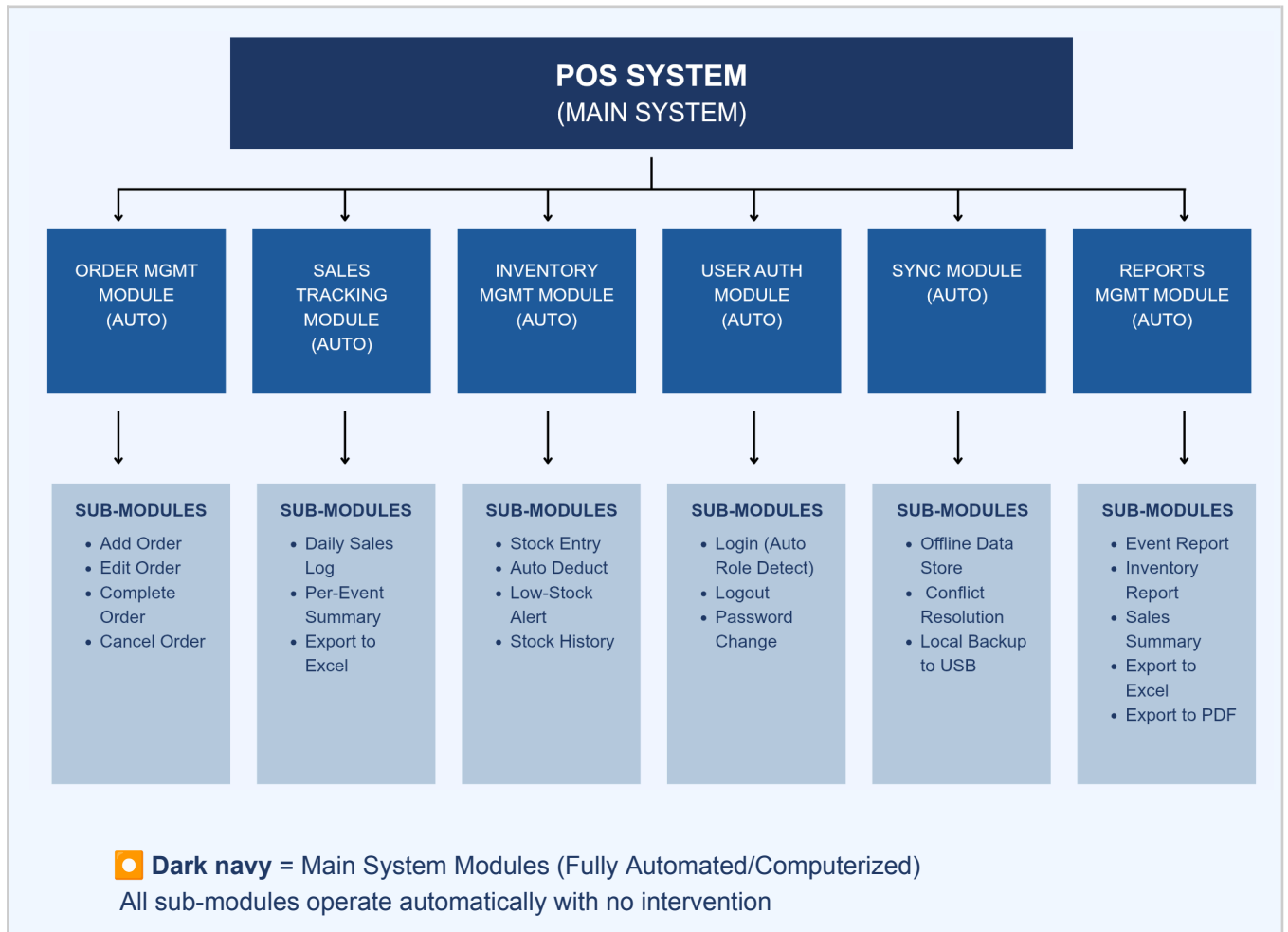
Modules and Sub-Modules

The following table and diagram describes each module of the proposed POS system:

Module	Sub-Modules	Description
Order Management	Add Order, Edit Order, Complete Order, Cancel Order	Allows staff to process customer orders on both laptop and tablet. Orders are saved locally and synced when internet is available.
Sales Tracking	Daily Sales Log, Per-Event Summary, Export to Excel	Automatically records each transaction. Replaces manual tallying. Generates per-event summaries exportable as .xlsx files.
Inventory Management	Stock Entry, Stock Deduction (per order), Low-Stock Alert	Monitors product stock levels in real time. Deducts from stock with each order. Alerts staff when items are running low.
User Authentication	Login, Logout, Role-Based Access	Secure staff login. Role-based access ensures only authorized users can view reports or modify inventory.
Sync Module	Offline Data Store, Sync-on-Connect, Conflict Resolution	Method 1 – Hotspot sync: After event, owner turns on laptop's WiFi hotspot. Tablets connect and send stored order data to laptop's MySQL database. Method 2 – USB transfer: After event, owner connects tablet to laptop via USB cable and copies the exported order file. Method 3 – Manual Excel file: Tablet exports orders as Excel (.xlsx) or CSV file to USB drive. Owner saves to laptop and imports to MySQL. Important: No data transfer happens during the event. All tablets work offline. Sync only occurs after the event ends. Conflict resolution: Owner manually reviews all data on laptop before final import. No automatic merging.

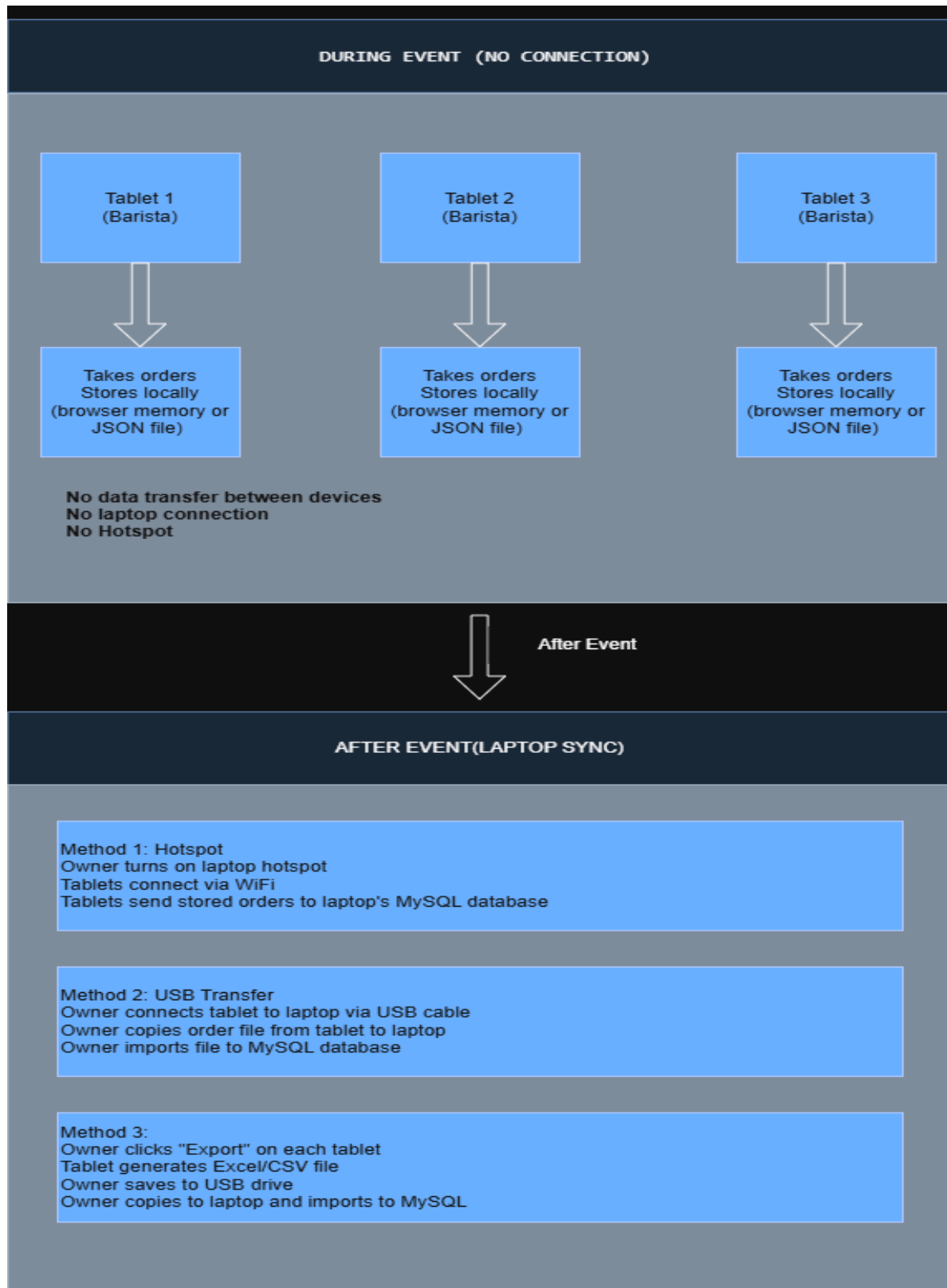


Module	Sub-Modules	Description
Reports Module	Event Report, Inventory Report, Excel Export	Generates post-event performance reports. All reports can be exported to Excel for record-keeping and analysis.





6.3.1. Multi-Device Workflow (Offline During Event, Sync After Event)





Time	Action
During event	Tablets work offline. Each tablet stores its own orders. No communication between devices.
After Event	Owner chooses one of three methods to transfer data from tablets to laptop.
After Import	Owner reviews all orders on laptop, generates reports, exports to Excel for backup.

6.3.2. Transaction Validation (On Tablet During Event)

Each order is validated on the tablet before being stored locally :

Validation Rule	How It Works
Order must have items	Cannot submit empty order.
Quantity is positive	Rejects zero or negative quantities
Product exists	Product must be in the loaded menu
Price is from system	Price is pulled from internal menu, not typed by barista
Timestamp recorded	Each order has date and time
Unique order number	Order ID = Tablet number + sequential number (e.g., T1-001, T2-001, T3-001)

How data is stored on tablet (temporarily):

Storage Method	Description
Browser memory	Orders are kept in JavaScript arrays while app is open
JSON file	Optional: Tablet can save orders as JSON file for backup



No database

No SQLite, no MySQL on tablet. Simple file-based storage.

Validation during laptop import (after event):

Check	Action if Failed
Duplicate order ID	Owner is alerted. Manually reviews and deletes duplicate.
Invalid date/time	Owner corrects before import.
Quantity mismatch	Owner reviews and adjusts if needed.
Missing product	Owner adds missing product to MySQL or skips the order.

6.3.3. Offline & Sync Error Handling

Scenario 1: Tablet loses stored orders during event (app crashes)

Step	Action
1	Barista restarts the POS app
2	If orders were saved as JSON file, they can be recovered
3	If not saved, barista re-enters orders from paper log (if available)
4	Prevention: Tablet auto-saves after every 5 orders



Scenario 2: Hotspot sync fails after event (Method 1)

Step	Action
1	Owner retries connection 3 times
2	If still fails, owner switches to Method 2 (USB) or Method 3 (Excel file)
3	No data is lost because orders are still stored on tablet

Scenario 3: Excel/CSV export fails (Method 3)

Step	Action
1	Tablet displays error message
2	Owner retries export
3	If still fails, owner manually copies the internal JSON file via USB
4	Owner opens JSON file and copies data to Excel manually

Scenario 4: Laptop import fails (file corrupted)

Step	Action
1	Laptop shows "Import failed" message
2	Owner opens the Excel/CSV file manually
3	Owner checks for obvious errors (empty rows, wrong columns)
4	Owner fixes file and retries import
5	If unfixable, owner re-exports from tablet



Scenario 5: Tablet storage gets full during event

Step	Action
1	Tablet shows warning at 80% storage
2	Owner exports current orders to USB drive mid-event
3	Owner clears exported orders from tablet
4	Tablet continues recording new orders

Scenario 6: Two tablets recorded the same order (customer ordered twice)

Step	Action
1	This is a business process issue, not a system error
2	During laptop review, owner spots duplicate by timestamp and amount
3	Owner manually deletes duplicate order before finalizing
4	Deleted order is logged as "MANUALLY REMOVED" for audit trail

6.4. Platform

6.4.1. Operating System: Windows 10/11 (Laptop); Android / iPadOS (Tablet)

6.4.2. Usage: Intranet (Local WiFi during events) + Optional Online Sync

The system runs on the laptop's local server (XAMPP/WAMP). The tablet accesses the system through a web browser over the local WiFi or mobile hotspot created during events. No permanent internet connection is required.



6.4.3. Software Specifications:

6.4.3.1. Programming Language:

Front-end:	HTML5, CSS3, JavaScript, and Tailwind
Back-end:	PHP 8.x
Database:	MySQL 8.x
Local Server:	XAMPP or WAMP (runs PHP + MySQL on laptop)

6.4.3.2. Other software tools:

Report Export:	PHP to Excel (.xlsx) via PhpSpreadsheet library
Browser (Client):	Google Chrome or Microsoft Edge (on Tablet)
Receipt Printing:	NOT INCLUDED – not required by the business
Version Control:	Git / GitHub (for development)
USB Drive:	Local backup storage (after each event)

6.4.4. Hardware Specifications (minimum specs):

6.4.4.1. Workstation (Laptop):

Processor:	Intel Core i3 (8th Gen) or equivalent
RAM:	4 GB minimum, 8 GB recommended
Storage:	128 GB SSD or 250 GB HDD
OS:	Windows 10 / 11 (64-bit)
Display:	13" or larger
Connectivity:	WiFi 802.11 b/g/n (to create hotspot for tablet)

6.4.4.2. Tablet (Client Device)

Device:	Any Android tablet (7"+) or iPad running a modern browser
Browser:	Google Chrome / Safari (latest version)
Connectivity:	WiFi (connects to laptop's hotspot or event WiFi)
Display:	7" minimum, 10" recommended for touch usability

6.4.4.3. Printer



No printer is required. Receipt printing is intentionally excluded from the system as TravelBean MNL clients do not request printed receipts. All records are stored digitally.

6.4.4.4. Network Infrastructure:

A. Bandwidth:	No minimum required during events (offline operation)
B. ISP:	NOT REQUIRED for operations. Owner may use mobile data only for optional email reports AFTER events.
C. Topology / Cabling:	Star topology using WiFi hotspot. No physical cabling.
D. Routers / Switches:	No dedicated router required. Laptop's built-in Windows Mobile Hotspot is used.
E. Others:	No barcode reader, biometric device, or RFID required for this system.

7. Additional Technical Specifications

7.1 Backup and Recovery Plan

Backup Type	Frequency	Location
Automatic .sql backup	Every 24 hours	Laptop secondary folder
Excel export	After each event	USB drive (kept separate from laptop)
Manual backup	Before any system update	USB drive + laptop folder

7.2 Recovery Procedures:

Scenario	Action
Laptop fails during event	Staff switches to temporary paper log. After laptop is restored, missed orders are manually re-entered into POS before next export.



Tablet failure	No data loss – tablet is only a client. Laptop holds all records. Staff can use laptop directly or switch to a backup tablet.
Database corruption	Restore from most recent .sql backup file (daily). Maximum data loss = 24 hours.
USB drive loss/damage	No critical impact – Excel export is for record-keeping only. Primary database remains on laptop.

7.3 User Roles and Access Rights

Role	Permissions
Barista / Staff	Add order, edit order, cancel order, complete order, view current inventory levels
Owner / Co-Owner	All staff permissions + view reports, modify inventory (stock entry/adjustments), export to Excel, manage USB backup

7.4 Conflict Resolution Policy

Record Type	Conflict Resolution Rule
Sales Records	Append-only – never overwritten or deleted
Edited Orders	Original values + new values + timestamp + staff ID logged separately
Cancelled Orders	Original order preserved with "CANCELLED" flag + timestamp



Inventory Edits

"Last write wins" based on most recent staff action

7.5.1 Data Security on Tablet (During Event)

Risk	Mitigation
Tablet theft	No customer payment data stored (cash only). Order data is temporary. No MySQL database on tablet.
Unauthorized access	POS app requires login.
Data tampering	Orders are stored as JSON or Excel/CSV. Owner reviews all files before import.

7.5.2 Data Security on Laptop (MySQL Database)

Risk	Mitigation
MySQL unauthorized access	MySQL has root password. Database user has only INSERT, SELECT, UPDATE permissions (no DELETE or DROP).
Laptop theft	Daily .sql backups. USB backup kept separately from laptop.
SQL injection	PHP uses prepared statements for all database queries.

7.5.3 User Authentication

User Type	Device Authentication Method
Barista Tablet	Username + password (hashed using PHP password_hash())
Owner/Co-owner	LaptopUsername + password (hashed) + Windows login



7.5.4 Excel/CSV File Security (For Manual Transfer)

Measure	Implementation
Password protection	Excel files can be encrypted with a password before transfer (optional)
Manual review	Owner always opens and reviews Excel file before importing to MySQL
USB encryption	USB drive can be encrypted if sensitive data is stored